

# FHRP — First Hop Redundancy Protocols

## CCNA Notes + Examples + Packet Flow

These notes combine the concepts from **all the images you sent**, including **FHRP, HSRP, VRRP, GLBP, ARP, Gratuitous ARP, failover, preemption, load balancing, and HSRP configuration**.

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## 1. What is FHRP?

**FHRP = First Hop Redundancy Protocol**

FHRP is used to provide **redundancy for the default gateway** of end devices.

Normally, a PC has only one default gateway:

PC

IP: 172.16.0.1

DG: 172.16.0.254

If the router providing **172.16.0.254** fails:

PC

|

| Default Gateway = 172.16.0.254

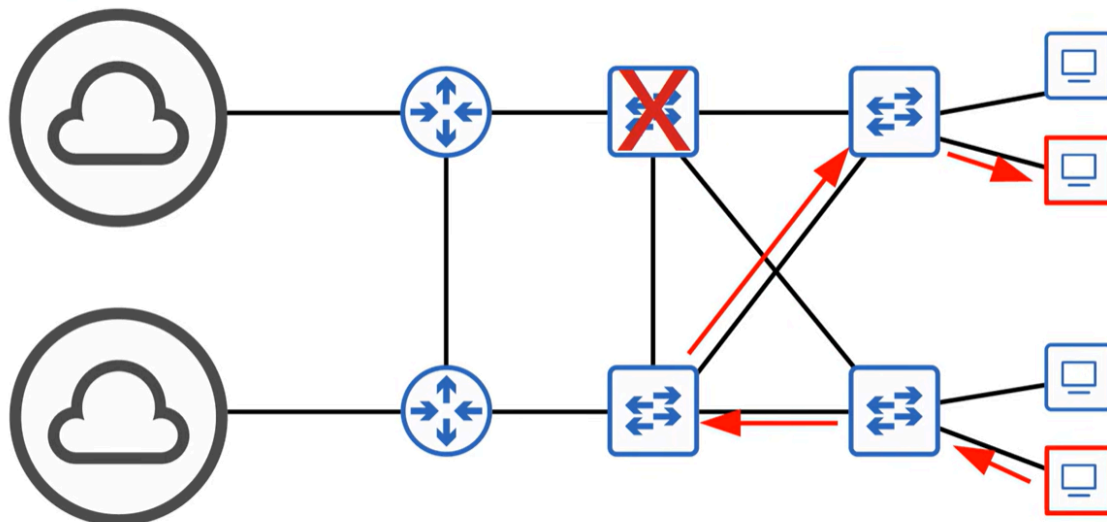
|

R1 

The PC cannot reach other networks, even if another router is available.



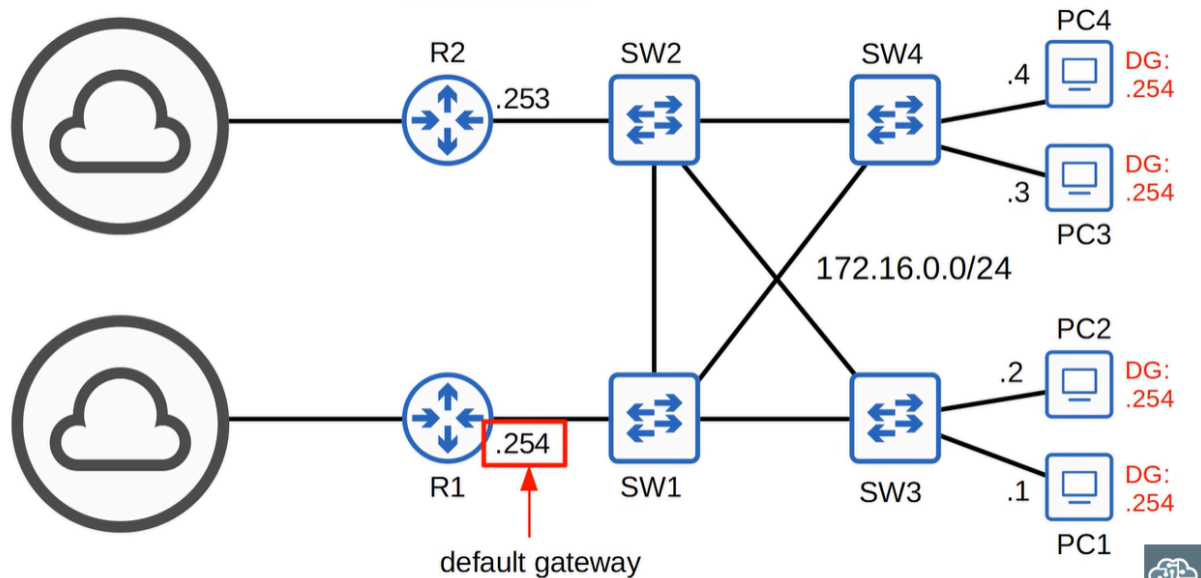
## Network Redundancy



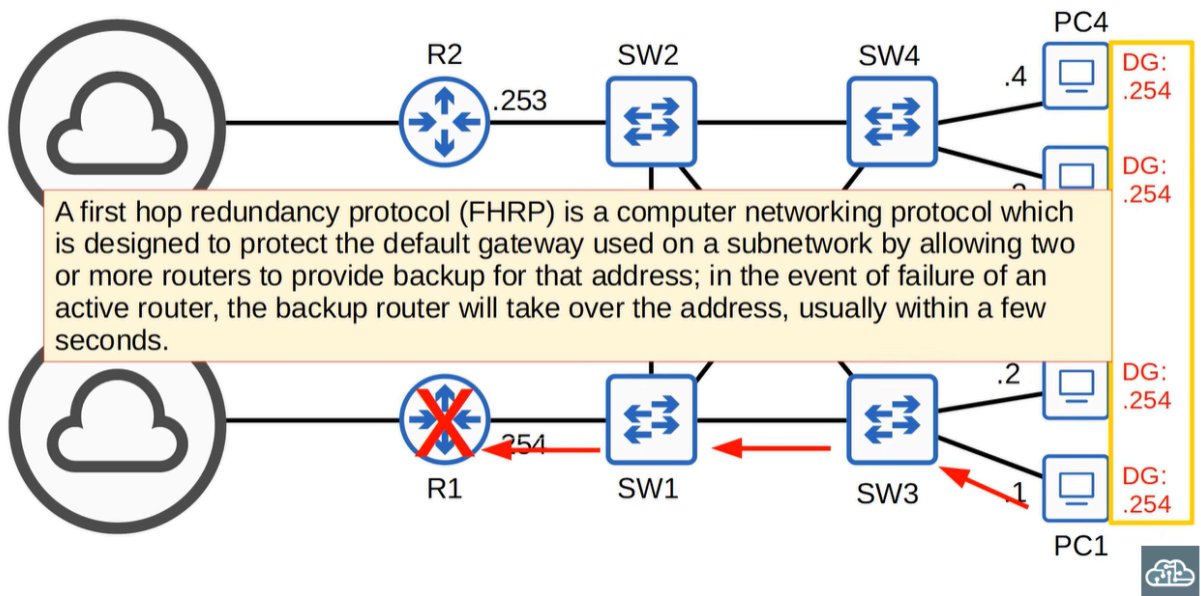
That's not a problem, because this alternate path is available.



## First Hop Redundancy Protocols



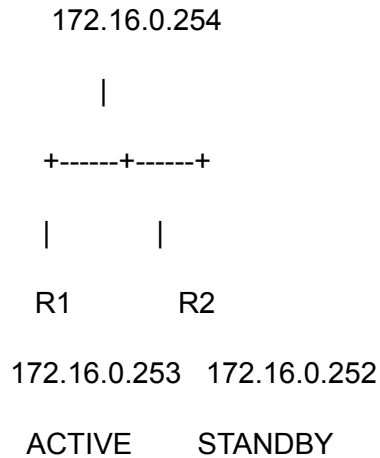
## First Hop Redundancy Protocols



**FHRP solves this problem.**

We have two routers:

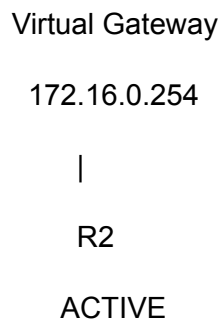
Virtual Gateway



The PCs use:

Default Gateway = 172.16.0.254

If R1 fails:



The PCs **do not need to change their default gateway.**

## 2. Why do we need a Virtual IP?

This is one of the most important FHRP concepts.

Suppose:

R1 = 172.16.0.253

R2 = 172.16.0.252

Without FHRP, the PC would have to choose one:

Default Gateway = 172.16.0.253

If R1 fails, the PC would need to be reconfigured to:

Default Gateway = 172.16.0.252

That's obviously not practical.

So FHRP creates:

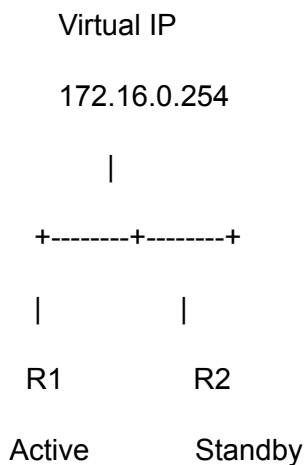
Virtual IP = 172.16.0.254

The PC uses:

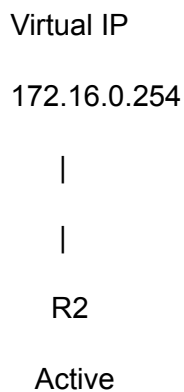
Default Gateway = 172.16.0.254

The virtual IP remains the same even when the physical active router changes.

**Think of it like this:**



After R1 fails:



The address stays the same. Only the router providing that address changes.

---

### 3. Virtual MAC Address

IP operates at **Layer 3**, but the PC must actually create an Ethernet frame at **Layer 2**.

Therefore, the PC needs a MAC address corresponding to the virtual gateway.

FHRP provides:

Virtual IP



Virtual MAC

Example:

Virtual IP = 172.16.0.254

Virtual MAC = 0000.0c9f.f001

The PC's ARP table might contain:

172.16.0.254 → 0000.0c9f.f001

Then the PC sends:

Source MAC = PC's MAC

Destination MAC = Virtual MAC

Source IP = 172.16.0.1

Destination IP = 8.8.8.8

Notice:

**The destination IP is still 8.8.8.8.**

The virtual IP is used as the **default gateway**, not as the final destination.

---

## 4. Physical IP vs Virtual IP

This distinction is extremely important.

Example:

R1 physical IP = 172.16.0.253

R2 physical IP = 172.16.0.252

Virtual IP = 172.16.0.254

PC:

IP Address = 172.16.0.1

Default Gateway = 172.16.0.254

The PC does **not** use:

172.16.0.253

or:

172.16.0.252

It uses:

172.16.0.254

---

## 5. FHRP Election

FHRP routers communicate with each other to determine who should provide the gateway.

Typically there is:

Active

Standby

or, depending on protocol:

Master

Backup

The terminology depends on the FHRP.

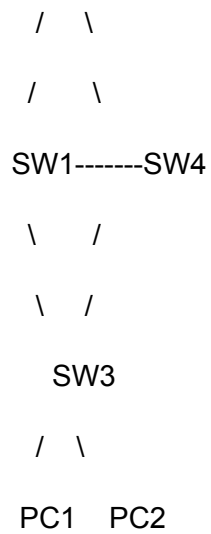
| Protocol | Primary router | Backup router          |
|----------|----------------|------------------------|
| HSRP     | Active         | Standby                |
| VRRP     | Master         | Backup                 |
| GLBP     | AVG / AVF      | Different architecture |

---

## 6. Example Network

Use this topology to understand the whole process:

```
Internet
|
R2
172.16.0.253
STANDBY
|
SW2
```



R1

172.16.0.254

ACTIVE

LAN:

172.16.0.0/24

Hosts:

PC1 = 172.16.0.1

PC2 = 172.16.0.2

PC3 = 172.16.0.3

PC4 = 172.16.0.4

Gateway:

172.16.0.254

Routers:

R1 = 172.16.0.254

R2 = 172.16.0.253

But in the FHRP configuration, **172.16.0.254** is the **VIP**, so the physical addresses would normally be distinct, e.g.:

R1 = 172.16.0.253

R2 = 172.16.0.252

VIP = 172.16.0.254

---

## 7. What happens when PC1 wants to reach 8.8.8.8?

PC1:

IP = 172.16.0.1

DG = 172.16.0.254

Destination:

8.8.8.8

PC1 realizes:

8.8.8.8 is outside my subnet.

Therefore:

"I need to send this packet to my default gateway."

But PC1 doesn't initially know the MAC address for:

172.16.0.254

So it sends an ARP request.

---

## 8. ARP Request

PC1 broadcasts:

Who has 172.16.0.254?

My IP is 172.16.0.1.

What is your MAC address?

Destination MAC:

FFFF.FFFF.FFFF

This is a Layer-2 broadcast.

The switch floods it throughout the local broadcast domain.

---

## 9. Active Router Replies

R1 is the **Active** router.

Therefore R1 replies:

172.16.0.254 is at

Virtual MAC

The PC stores:

ARP Table

172.16.0.254

↓

Virtual MAC

Important:

**R1 doesn't necessarily reply with its physical MAC.**

It replies using the **FHRP virtual MAC**.

---

## 10. PC Sends the Packet

Now PC1 creates:

Ethernet Frame:

Source MAC:

PC1's MAC

Destination MAC:

Virtual MAC

Inside it:

IP Packet:

Source IP:

172.16.0.1

Destination IP:

8.8.8.8

So:

PC1

|

| Destination MAC = Virtual MAC

| Destination IP = 8.8.8.8

↓

R1

R1 receives the frame because R1 is currently the active owner of the virtual gateway.

R1 routes the packet toward the Internet.

---

## 11. What happens if R1 fails?

Before failure:

R1 = ACTIVE

R2 = STANDBY

VIP = 172.16.0.254

V-MAC = Virtual MAC

R1 stops sending its FHRP **Hello messages**.

R2 notices:

"I haven't received Hello messages from R1."

Therefore R2 becomes:

ACTIVE

Now:

R1 ❌

R2 = ACTIVE

VIP = 172.16.0.254

V-MAC = same virtual MAC

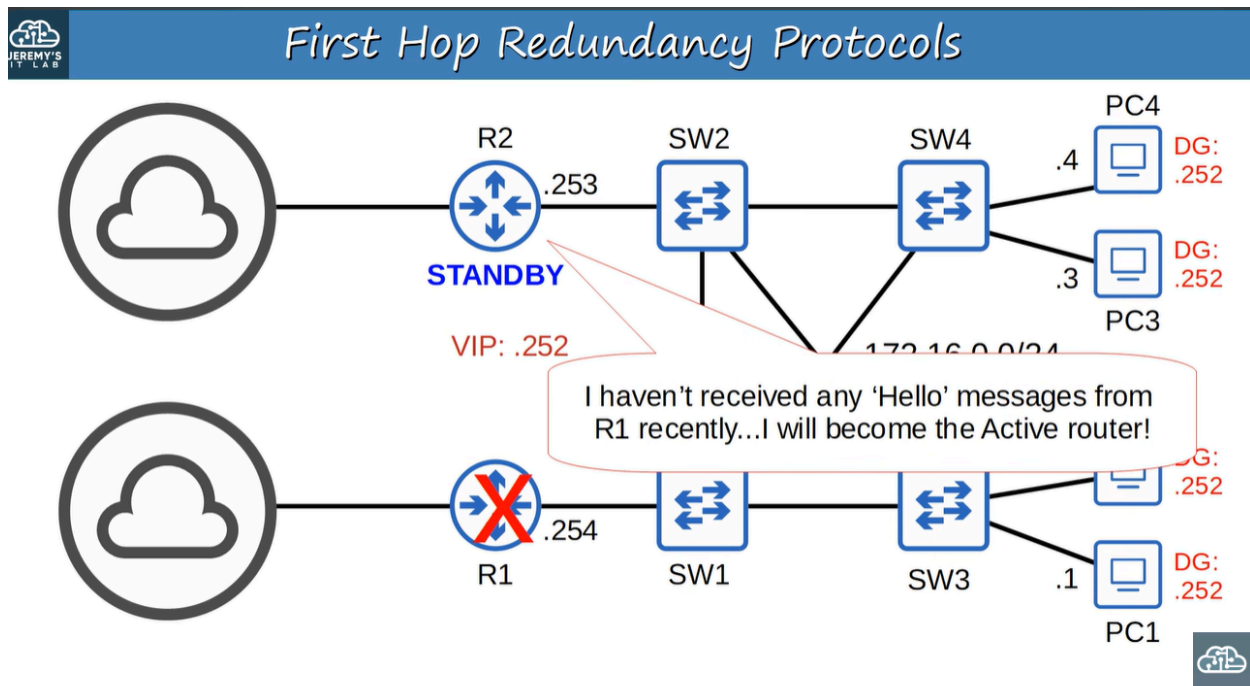
### Most important point:

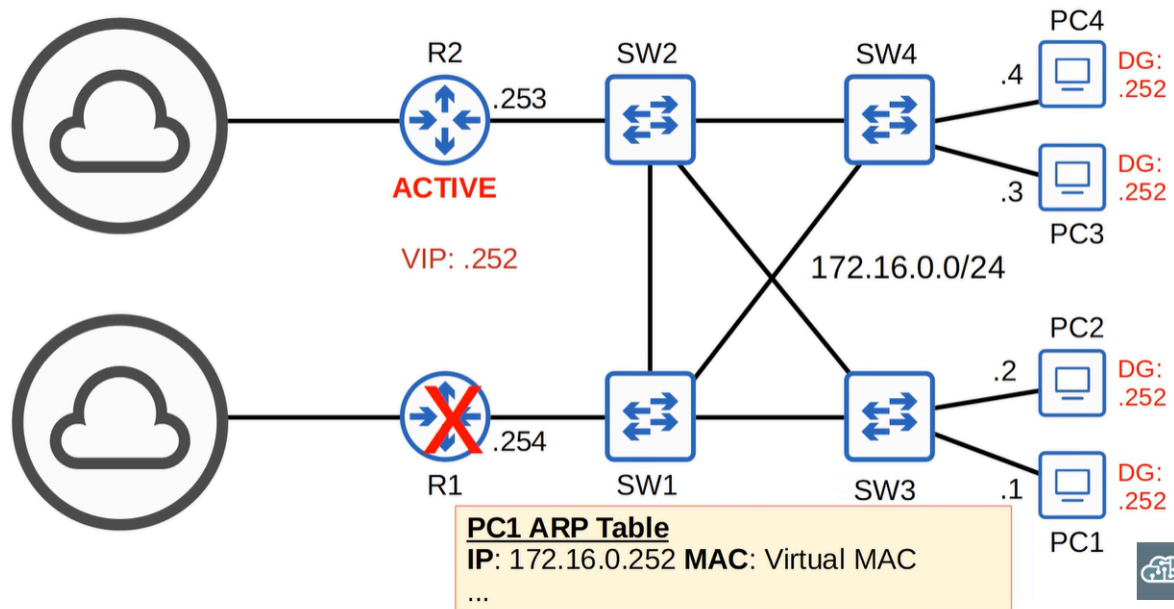
The PC's gateway does NOT change.

Still:

172.16.0.254

And the virtual MAC remains the same.





## 12. Why Gratuitous ARP?

This connects directly to your earlier question.

After R2 becomes active, it sends **Gratuitous ARP (GARP)**.

A normal ARP is:

"Who has this IP?"

A Gratuitous ARP is essentially an unsolicited announcement:

"Hello!

172.16.0.254 is associated with my virtual MAC.

I'm now the active router."

It is sent without first receiving an ARP request.

## **Main purpose in FHRP:**

It helps switches quickly update their MAC address tables.

For example, before failure:

Virtual MAC → Switch Port toward R1

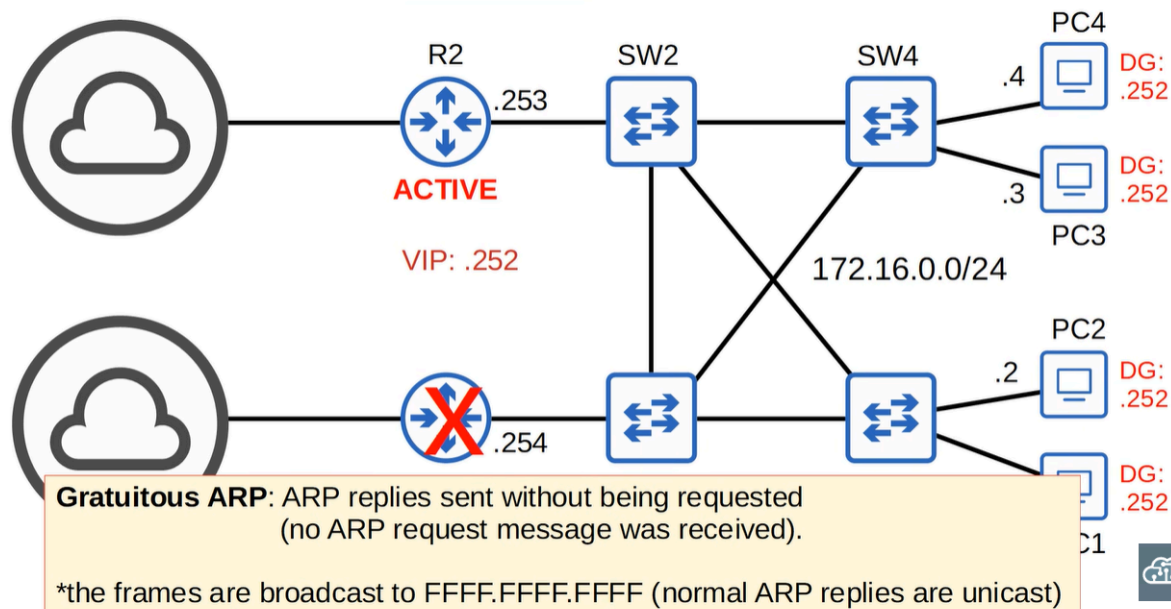
After R1 fails:

Virtual MAC → Switch Port toward R2

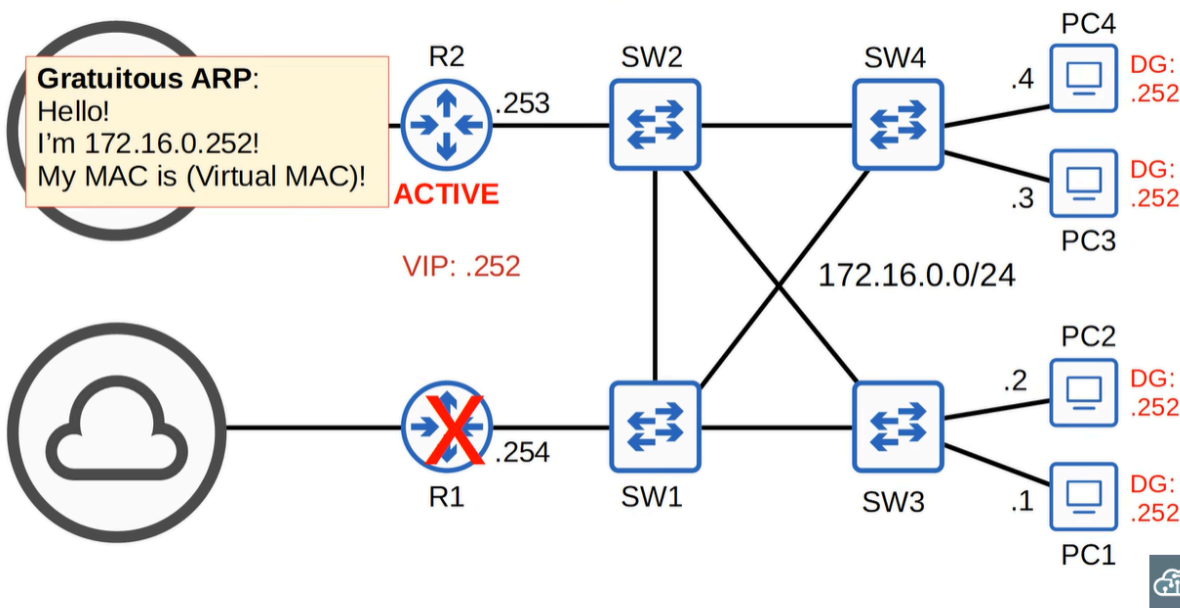
The switch needs to learn that the virtual MAC is now reachable through the interface toward R2.

GARP helps trigger that relearning quickly.

## First Hop Redundancy Protocols



## First Hop Redundancy Protocols



# 13. Important: GARP is not the mechanism that elects the new router

This distinction is important.

## **FHRP Hello messages:**

Used for:

Neighbor communication

Health detection

Election/state maintenance

## **Gratuitous ARP:**

Used after takeover to help:

Update/relearn MAC address information

So:

Hello messages

↓

R1 failure detected

↓

R2 becomes Active

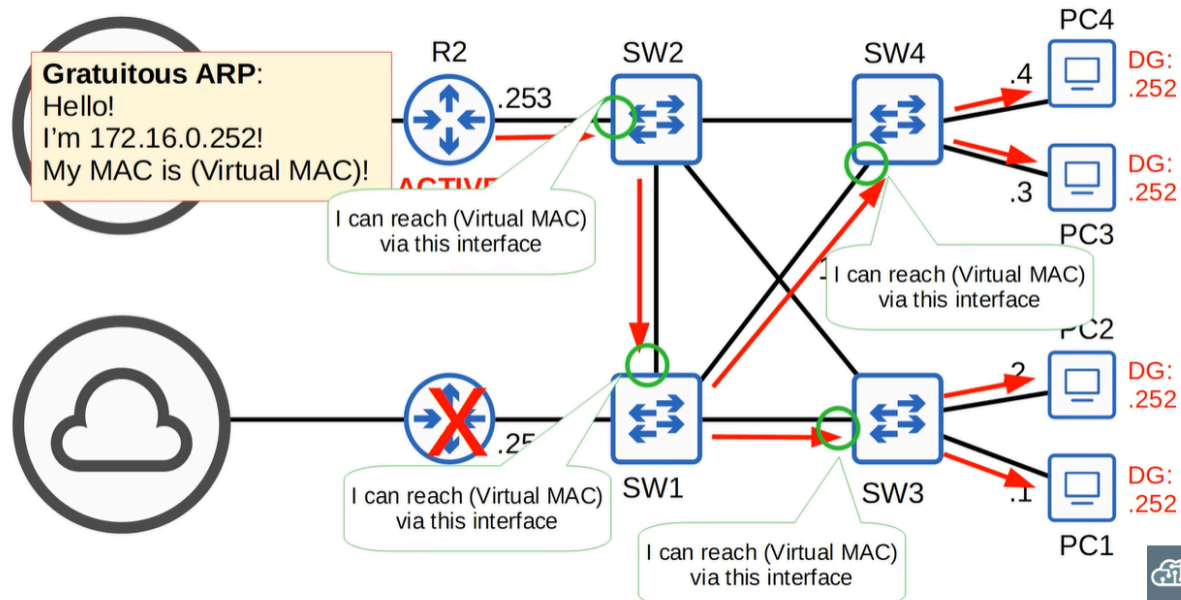
↓

Gratuitous ARP

↓

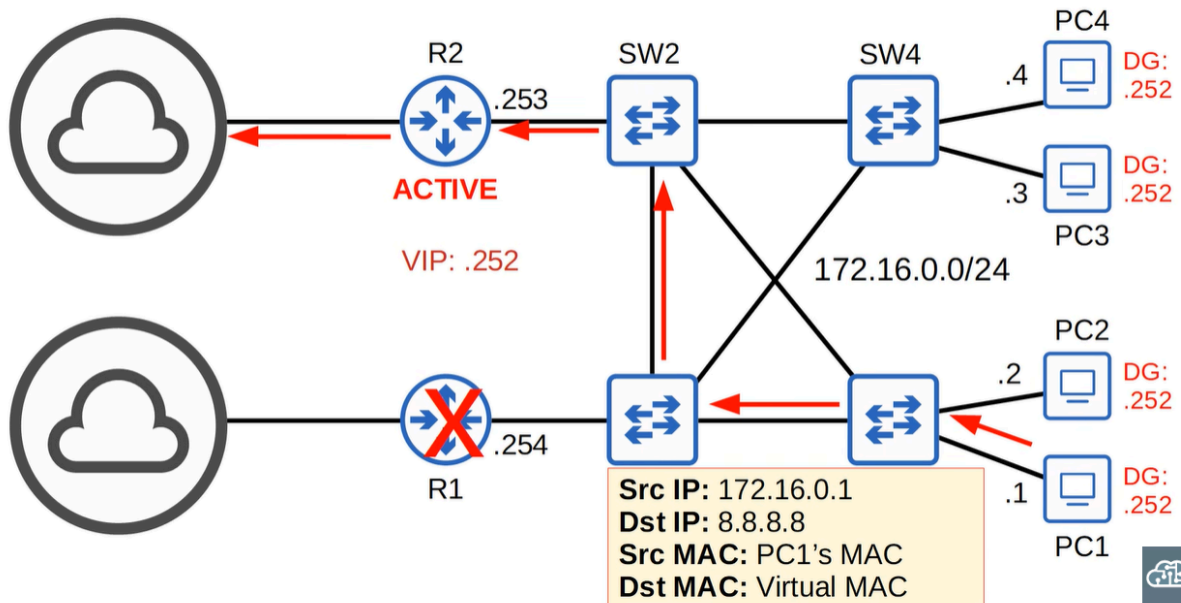
Switches relearn Virtual MAC location

# First Hop Redundancy Protocols



Traffic continues

# First Hop Redundancy Protocols



## 14. What happens to the PC's ARP table?

Suppose PC1 already knows:

172.16.0.254 → Virtual MAC

R1 fails.

The mapping can remain:

172.16.0.254 → same Virtual MAC

That's the beauty of FHRP.

The PC doesn't have to learn:

172.16.0.252

for R2.

Instead:

Virtual IP

+

Virtual MAC

↓

New Active Router

The **identity remains the same.**

---

## 15. FHRP Non-Preemptive Behavior

By default, FHRP is generally **non-preemptive**.

Example:

Initially:

R1 = Active

R2 = Standby

R1 fails:

R1 ✗

R2 = Active

Later R1 comes back:

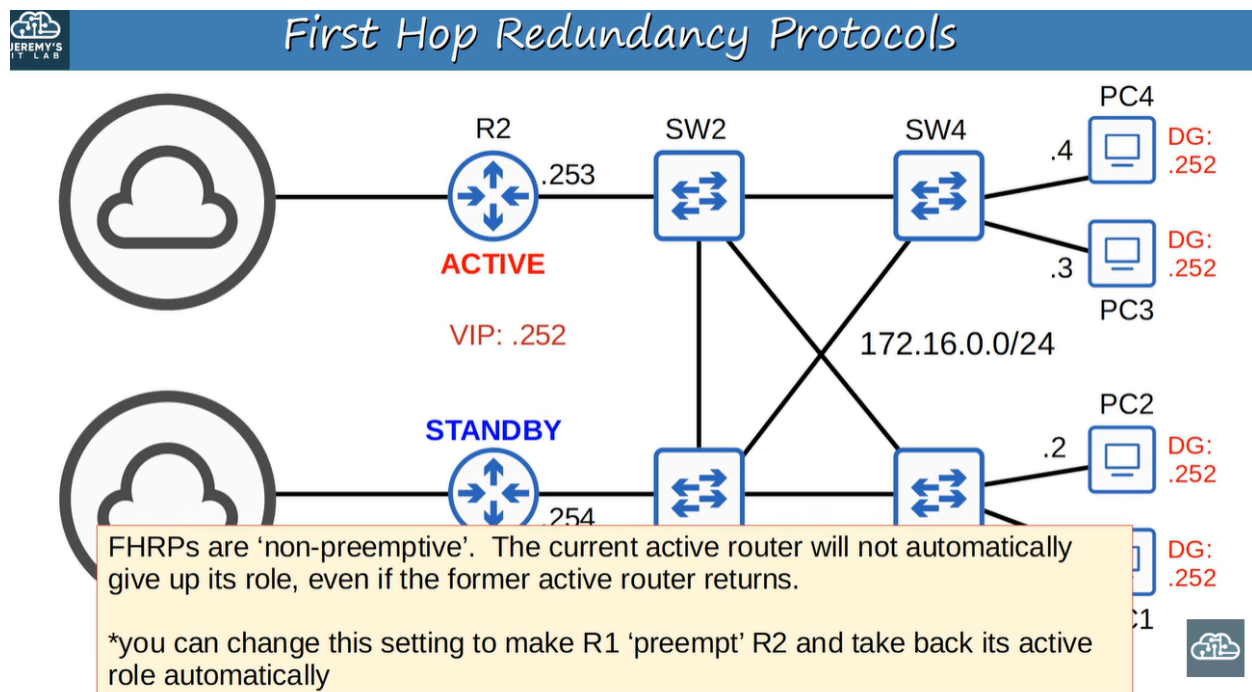
R1 = Standby

R2 = Active

R1 does **not automatically take back Active status** just because it has returned.

Why?

Because R2 is already successfully serving as the gateway.



## 16. Preemption

You can configure **preemption**.

Then:

R1 = higher priority

R2 = lower priority

Initially:

R1 = Active

R2 = Standby

R1 fails:

R2 = Active

R1 returns:

R1 has higher priority

↓

Preemption

↓

R1 becomes Active again

R2 becomes Standby

**HSRP command:**

```
interface g0/0
```

```
standby 1 preempt
```



## First Hop Redundancy Protocols

- A **virtual IP** is configured on the two routers, and a **virtual MAC** is generated for the virtual IP (each FHRP uses a different format for the virtual MAC)
  - An **active** router and a **standby** router are elected. (different FHRPs use different terms)
  - End hosts in the network are configured to use the virtual IP as their default gateway.
  - The active router replies to ARP requests using the virtual MAC address, so traffic destined for other networks will be sent to it.
  - If the active router fails, the standby becomes the next active router. The new active router will send **gratuitous ARP** messages so that switches will update their MAC address tables. It now functions as the default gateway.
  - If the old active router comes back online, by default it won't take back its role as the active router. It will become the standby router.
  - You can configure 'preemption', so that the old active router does take back its old role. 
- 

## 17. HSRP

### HSRP = Hot Standby Router Protocol

HSRP is:

- **Cisco proprietary**
- Uses **Active/Standby**
- Provides default gateway redundancy
- Supports HSRP groups
- Has version 1 and version 2



## HSRP (Hot Standby Router Protocol)

- Cisco proprietary.
- An **active** and **standby** router are elected.
- There are two versions: **version 1** and **version 2**.  
Version 2 adds IPv6 support and increases the number of *groups* that can be configured.
- Multicast IPv4 address: v1 = 224.0.0.2  
v2 = 224.0.0.102 **0000.0c07.ac01**
- Virtual MAC address: v1 = 0000.0c07.acXX (XX = HSRP group number)  
v2 = 0000.0c9f.fXXX (XXX = HSRP group number) **0000.0c9f.f001**
- In a situation with multiple subnets/VLANs, you can configure a different active router in each subnet/VLAN to load balance.



## 18. HSRP Terminology

R1 → Active

R2 → Standby

The active router forwards traffic.

The standby router waits and monitors the active router.

## 19. HSRP Version 1 vs Version 2

| Feature | HSRP v1 | HSRP v2 |
|---------|---------|---------|
| IPv4    | Yes     | Yes     |
| IPv6    | No      | Yes     |

|                |                    |                    |
|----------------|--------------------|--------------------|
| Multicast IPv4 | 224.0.0.2          | 224.0.0.102        |
| Virtual MAC    | 0000.0c07.a<br>cXX | 0000.0c9f.f<br>XXX |
| Groups         | Fewer              | More groups        |

### HSRP v1 multicast:

224.0.0.2

### HSRP v2 multicast:

224.0.0.102

---

## 20. HSRP Virtual MAC

### HSRP Version 1

0000.0c07.acXX

XX represents the HSRP group number.

Example:

Group = 1

Virtual MAC =

0000.0c07.ac01

### HSRP Version 2

0000.0c9f.fXXX

Example:

Group = 1

Virtual MAC =

0000.0c9f.f001

---

## 21. HSRP Hello Messages

HSRP routers periodically exchange Hello messages.

Example:

R1

|

| Hello

↓

R2

This allows R2 to know:

R1 is still alive.

If the Hellos stop arriving:

R1 

↓

R2 detects failure

↓

R2 becomes Active

Typical HSRP timers shown in your slides:

Hello = 3 seconds

Hold = 10 seconds

---

## 22. HSRP Priority

HSRP uses **priority** to determine which router should become Active.

Default priority:

100

Higher priority wins.

Example:

R1 priority = 200

R2 priority = 50

Therefore:

R1 → Active

R2 → Standby

If priorities are equal, the higher IP address is used as the tie-breaker.

### **Election order:**

1. Highest priority
  2. Highest IP address
- 

## 23. HSRP Configuration

Example:

Network = 172.16.0.0/24

R1 = 172.16.0.253

R2 = 172.16.0.252

VIP = 172.16.0.254

## **R1**

```
R1(config)# interface g0/0
```

```
R1(config-if)# standby version 2
```

```
R1(config-if)# standby 1 ip 172.16.0.254
```

```
R1(config-if)# standby 1 priority 200
```

```
R1(config-if)# standby 1 preempt
```

## **R2**

```
R2(config)# interface g0/0
```

```
R2(config-if)# standby version 2
```

```
R2(config-if)# standby 1 ip 172.16.0.254
```

```
R2(config-if)# standby 1 priority 50
```

```
R2(config-if)# standby 1 preempt
```

Result:

R1 → Active

R2 → Standby

VIP → 172.16.0.254

## **Important:**

HSRP v1 and v2 are **not compatible with each other**.

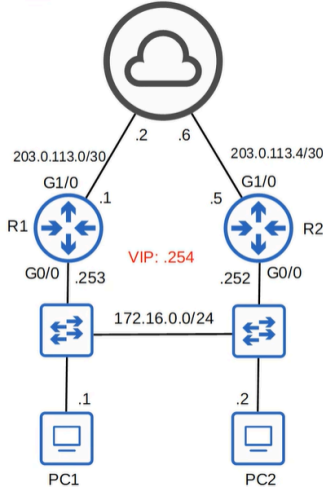
If R1 uses:

standby version 2

R2 must also use:

standby version 2

## 24. HSRP Configuration Commands — Meaning



```
R1(config)#interface g0/0
R1(config-if)#standby ?
<0-255>      group number
authentication Authentication
bfd          Enable HSRP BFD
ip           Enable HSRP IPv4 and set the virtual IP address
ip6          Enable HSRP IPv6
mac-address  Virtual MAC address
name         Redundancy name string
preempt      Overthrow lower priority Active routers
priority     Priority level
timers       Hello and hold timers
track        Priority tracking
```

### Configure virtual IP

standby 1 ip 172.16.0.254

Means:

HSRP Group 1

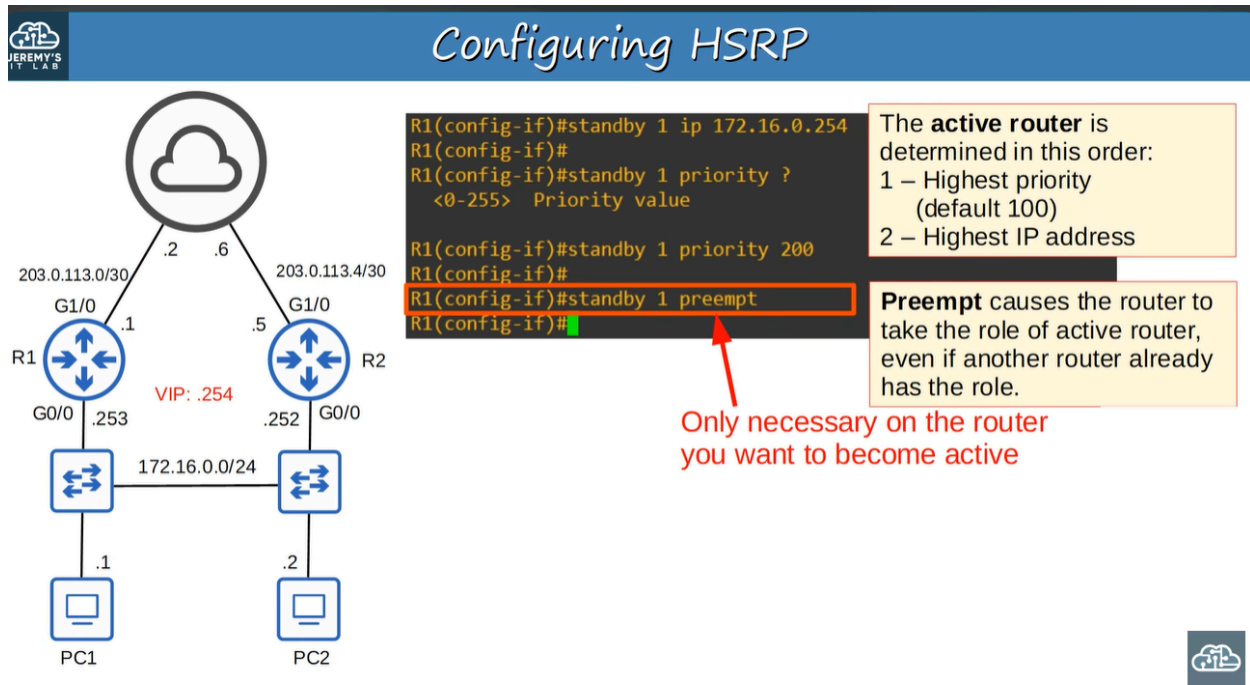
Virtual IP = 172.16.0.254

### Configure priority

standby 1 priority 200

Means:

HSRP priority = 200



## Enable preemption

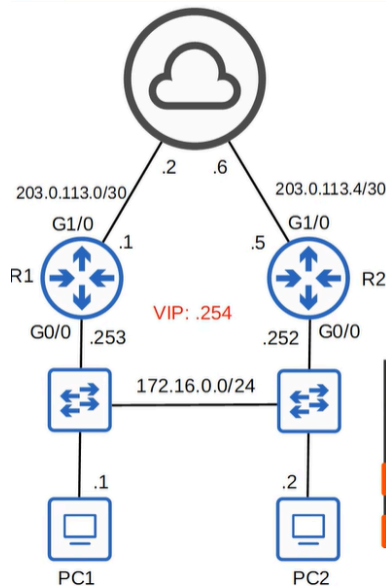
standby 1 preempt

Means:

If this router has higher priority, it can take back the Active role when it returns.



## Configuring HSRP



```
R1(config-if)#standby 1 ip 172.16.0.254
R1(config-if)#
R1(config-if)#standby 1 priority ?
<0-255> Priority value

R1(config-if)#standby 1 priority 200
R1(config-if)#
R1(config-if)#standby 1 preempt
R1(config-if)#
```

```
R2(config-if)#standby version 2
R2(config-if)#
R2(config-if)#standby 1 ip 172.16.0.254
R2(config-if)#
R2(config-if)#standby 1 priority 50
R2(config-if)#
R2(config-if)#standby 1 preempt
```

HSRP version 1 and version 2 are not compatible. If R1 uses version 2, R2 must use version 2 also.



### Change HSRP version

standby version 2

## 25. Verify HSRP

Command:

show standby

Example output:

State is Active

Virtual IP address is 172.16.0.254

Active virtual MAC address is

0000.0c9f.f001

Hello time 3 sec

hold time 10 sec

Preemption enabled

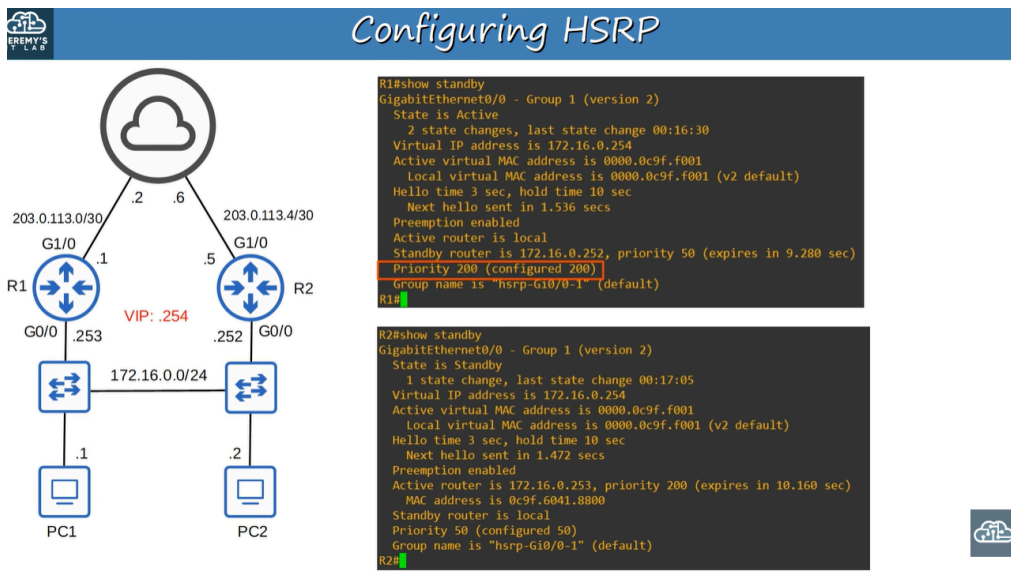
Standby router is 172.16.0.252

priority 50

Priority 200

From this output you can identify:

- Active/Standby state
- Virtual IP
- Virtual MAC
- Hello timer
- Hold timer
- Preemption
- Standby router
- Priority



---

## 26. Load Balancing with Multiple VLANs

HSRP itself has one Active router per HSRP group.

But if you have multiple VLANs, you can distribute the Active role.

Example:

### **VLAN 10**

R1 = Active

R2 = Standby

### **VLAN 20**

R2 = Active

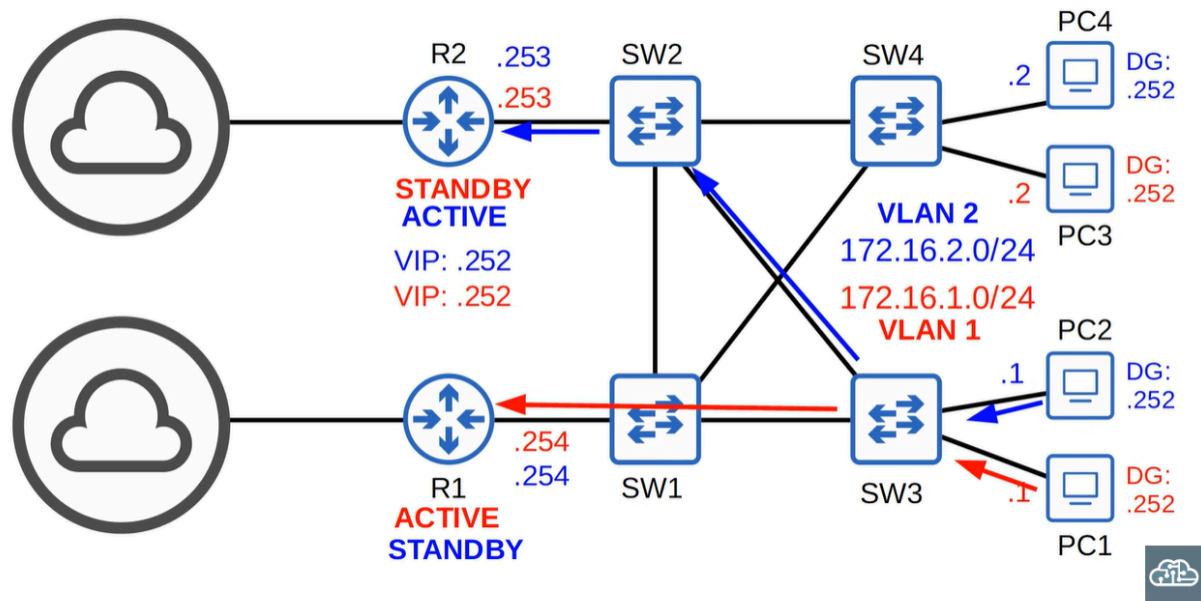
R1 = Standby

Therefore:

VLAN 10 traffic → R1

VLAN 20 traffic → R2

This provides a form of load distribution across VLANs.



## 27. VRRP

### VRRP = Virtual Router Redundancy Protocol

VRRP is an **open standard**.

Unlike HSRP:

HSRP → Cisco proprietary

VRRP → Open standard

Terminology:

Master

Backup

Not:

Active

Standby



## VRRP (Virtual Router Redundancy Protocol)



- Open standard
- A **master** and **backup** router are elected.
- Multicast IPv4 address: 224.0.0.18
- Virtual MAC address: 0000.5e00.01XX (XX = VRRP group number)

**0000.5e00.01c8**

(0xc8 = 200)

- In a situation with multiple subnets/VLANs, you can configure a different master router in each subnet/VLAN to load balance.

---

## 28. VRRP Example

R1 = 172.16.0.254

R2 = 172.16.0.253

VIP = 172.16.0.252

The PCs use:

Default Gateway = 172.16.0.252

Then:

R1 → Master

R2 → Backup

If R1 fails:

R2 → Master

The gateway remains:



JEREMY'S  
IT LAB

The diagram illustrates a network topology with two cloud providers (represented by cloud icons) connected to a core network. The core network consists of four switches (SW1, SW2, SW3, SW4) and two routers (R1, R2). R1 is labeled "MASTER BACKUP" and R2 is labeled "BACKUP MASTER".

Key components and connections:

- Routers:** R1 (MASTER BACKUP) and R2 (BACKUP MASTER) are connected to the cloud providers. R1 has a VIP of .252 and a backup VIP of .254. R2 has a backup VIP of .253 and a VIP of .252.
- Switches:** SW1, SW2, SW3, and SW4 are interconnected. SW1 and SW2 are connected to R1 and R2 respectively. SW2 and SW4 are connected to each other. SW1 and SW3 are connected to each other. SW2 and SW3 are connected to each other.
- VLANs:** VLAN 1 (172.16.1.0/24) and VLAN 2 (172.16.2.0/24) are shown. The default gateway (DG) for both VLANs is .252.
- PCs:** PC1, PC2, PC3, and PC4 are connected to the switches. PC1 and PC2 are connected to SW3. PC3 and PC4 are connected to SW4.

## 224.0.0.18

Group = 200

200 decimal = C8 hexadecimal

Virtual MAC:

0000.5e00.01C8

That's why the slide shows:

0000.5e00.01c8

---

## 31. VRRP Terminology

Remember:

HSRP

Active / Standby

VRRP

Master / Backup

This is a common exam question.

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## 32. GLBP

**GLBP = Gateway Load Balancing Protocol**

GLBP is:

- Cisco proprietary
- Provides gateway redundancy
- Provides **load balancing**

- Can use multiple routers simultaneously within a subnet

This is the major difference from traditional HSRP/VRRP operation.



## GLBP (Gateway Load Balancing Protocol)

- Cisco proprietary
- Load balances among multiple routers within a single subnet
- An **AVG (Active Virtual Gateway)** is elected.
- Up to four **AVFs (Active Virtual Forwarders)** are assigned by the AVG (the AVG itself can be an AVF, too)
- Each AVF acts as the default gateway for a portion of the hosts in the subnet.
- Multicast IPv4 address: 224.0.0.102
- Virtual MAC address: 0007.b400.XXYY (**XX** = GLBP group number, **YY** = AVF number)

0007.b400.0101



---

## 33. GLBP Terminology

GLBP uses:

### AVG

#### Active Virtual Gateway

The AVG controls the GLBP group and assigns forwarding responsibilities.

### AVF

#### Active Virtual Forwarder

An AVF actually forwards traffic.

The slide shows:

Up to four AVFs can be assigned by the AVG.

The AVG itself can also act as an AVF.

---

## 34. GLBP Example

Suppose:

R1

R2

R3

all participate in GLBP.

Instead of:

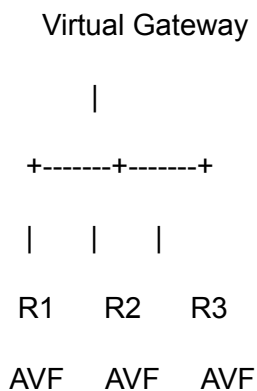
R1 = Active

R2 = Standby

R3 = waiting

GLBP can distribute hosts' default-gateway traffic among multiple routers.

Conceptually:



Different hosts can be directed toward different AVFs.

Therefore:

PC1 → R1

PC2 → R2

PC3 → R3

while still using the same virtual gateway IP.

---

## 35. GLBP Virtual MAC

Format:

0007.b400.XXYY

Where:

XX = GLBP group number

YY = AVF number

Example from the slide:

0007.b400.0101

Conceptually:

01 = group

01 = AVF

---

## 36. GLBP Multicast

GLBP IPv4 multicast:

224.0.0.102

Notice this is also the multicast address shown for **HSRP version 2**.

Don't confuse:

HSRP v2 → 224.0.0.102

GLBP → 224.0.0.102

They are different protocols even though they use the same multicast IPv4 address in this context.

---

## 37. FHRP Comparison

| Feature                | HSRP                        | VRRP                               | GLBP                            |
|------------------------|-----------------------------|------------------------------------|---------------------------------|
| Full name              | Hot Standby Router Protocol | Virtual Router Redundancy Protocol | Gateway Load Balancing Protocol |
| Vendor                 | Cisco                       | Open standard                      | Cisco                           |
| Primary terminology    | Active                      | Master                             | AVG                             |
| Backup terminology     | Standby                     | Backup                             | AVF architecture                |
| Redundancy             | Yes                         | Yes                                | Yes                             |
| Gateway load balancing | Not inherently              | Not inherently                     | <b>Yes</b>                      |
| IPv4 multicast         | v1: 224.0.0.2               | 224.0.0.18                         | 224.0.0.102                     |

|                   |     |    |     |
|-------------------|-----|----|-----|
| Cisco proprietary | Yes | No | Yes |
|-------------------|-----|----|-----|

## 38. The Complete FHRP Packet Flow

## Step 1 — PC needs another network

Therefore:

Send to default gateway.

---

## Step 2 — PC checks ARP

PC needs the MAC corresponding to:

172.16.0.254

If not in its ARP table:

ARP Request

"Who has 172.16.0.254?"

---

## Step 3 — Active router responds

R1 = Active

Response:

172.16.0.254

↓

Virtual MAC

PC stores:

ARP:

172.16.0.254

↓

Virtual MAC

---

## Step 4 — PC creates frame

Ethernet:

Src MAC = PC MAC

Dst MAC = Virtual MAC

IP:

Src IP = 172.16.0.1

Dst IP = 8.8.8.8

---

### Step 5 — Switch forwards the frame

The switch examines:

Destination MAC = Virtual MAC

and forwards the frame toward the active router.

---

### Step 6 — Active router routes it

R1

↓

Internet

↓

8.8.8.8

---

## 39. Now R1 Fails

R1 ❌

R2 is standby.

R2 detects the failure through the FHRP protocol.

Then:

R2 → Active

The virtual gateway remains:

VIP = 172.16.0.254

Virtual MAC remains:

same Virtual MAC

---

## 40. Gratuitous ARP After Failover

R2 sends:

Gratuitous ARP

Essentially:

"172.16.0.254 is here.

This is the virtual MAC."

Switches learn:

Virtual MAC

↓

Interface toward R2

Now traffic can flow:

PC

↓

Virtual MAC

↓

Switch

↓

R2

↓

Internet

---

## 41. The Most Important Mental Model

Don't think:

"The PC is sending to R1."

Think:

**"The PC is sending to the virtual gateway."**

Then FHRP decides:

Virtual Gateway

↓

Who currently owns it?

↓

Active router

Initially:

Virtual Gateway

↓

R1

After failure:

Virtual Gateway

↓

R2

The PC doesn't need to know that the physical router changed.

---

## 42. Three Addresses You Must Keep Separate

Suppose:

R1 physical IP = 172.16.0.253

R2 physical IP = 172.16.0.252

VIP = 172.16.0.254

There are three different IP addresses:

172.16.0.253 → R1's real IP

172.16.0.252 → R2's real IP

172.16.0.254 → FHRP virtual IP

And there are also:

R1 physical MAC

R2 physical MAC

Virtual MAC

The PC uses:

Default Gateway = Virtual IP

ARP → Virtual MAC

---

## 43. Very Important Correction

When we say:

"The virtual IP is configured on both routers"

don't interpret that as:

R1 interface physically owns VIP

AND

R2 interface physically owns VIP

That would create an IP conflict if treated as ordinary interface addressing.

Instead, both routers participate in the **same FHRP group**, and the FHRP protocol provides the virtual gateway identity.

Only the appropriate **Active/Master** router actually forwards traffic for that virtual gateway at a given time.

---

## 44. HSRP Example — Easy to Remember

Network: 192.168.1.0/24

R1 = 192.168.1.2

R2 = 192.168.1.3

VIP = 192.168.1.1

R1 priority = 200

R2 priority = 100

Therefore:

R1 → Active

R2 → Standby

PC:

IP = 192.168.1.10

DG = 192.168.1.1

If R1 fails:

R2 → Active

PC remains:

DG = 192.168.1.1

No PC reconfiguration is necessary.

---

## 45. HSRP Exam/Interview Points

Remember these:

### **FHRP**

Provides default gateway redundancy.

### **HSRP**

Cisco proprietary; Active/Standby.

### **VRRP**

Open standard; Master/Backup.

### **GLBP**

Cisco proprietary; gateway load balancing.

### **Virtual IP**

Used by end hosts as the default gateway.

## **Virtual MAC**

Used by hosts in the Ethernet frame to reach the virtual gateway.

## **Hello**

Used by FHRP routers to communicate and detect availability.

## **Gratuitous ARP**

Helps update/relearn Layer-2 MAC information after a gateway takeover.

## **Preemption**

Allows a higher-priority router to take back the Active role when it returns.

## **HSRP default priority**

100

## **HSRP election**

Highest priority

↓

Highest IP address if tied

## **HSRP v1 multicast**

224.0.0.2

## **HSRP v2 multicast**

224.0.0.102

## **VRRP multicast**

224.0.0.18

## **GLBP multicast**

224.0.0.102

---

# 46. One-Line Memory Trick

FHRP

↓

Protect the Gateway

HSRP

↓

Active / Standby

Cisco

VRRP

↓

Master / Backup

Open Standard

GLBP

↓

AVG / AVF

Load Balancing

Cisco

And the complete idea:

FHRP

|

Virtual Gateway

/

\

VIP

MAC

|

|

+-----+-----+

|

Active Router

|

Packet Forwarding

Active fails

↓

Standby/Master Backup

↓

Takes over VIP

↓

Gratuitous ARP

↓

Switches relearn MAC

↓

Traffic continues

This is the core FHRP concept you should be able to explain in an interview or CCNA exam.